SIMATS ENGINEERING
ASSESSMENT TOOL 3
COURSE NAME: Artificial Intelligence
COURSE CODE: CSA17

Code of Conduct:

I, **B.Vishnu** / **192512410** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ________________________

ANNEXURE A
ARTICLE REVIEW - SOLUTIONS

Task 1: Article Summary

(a) Article Citation and Domain

Citation: Liu, T., Tan, X., Sun, B., Wu, Y., & Guan, X. (2020). Energy Management of Cooperative Microgrids: A Multi-Agent Deep Reinforcement Learning Approach. *IEEE Transactions on Smart Grid*, 11(5), 4152-4162. DOI: 10.1109/TSG.2020.2985651.

Domain/Application Area: AI-driven Renewable Energy Optimization and Smart Grid Management.

Specific ML Problem Addressed: Continuous state-action space optimization using Reinforcement Learning for dynamic energy scheduling under high environmental uncertainty.

(b) Key Objective and Research Gap

The primary objective of this article is to design a decentralized, automated energy management framework for multiple cooperative microgrids integrated with renewable energy sources (solar and wind). The authors identified a significant research gap: traditional optimization methods (like dynamic programming or mixed-integer linear programming) require exact mathematical models of the environment and struggle computationally with the highly stochastic nature of renewable generation. To fill this gap, the authors proposed a model-free Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithm. This allows the microgrids (acting as RL agents) to learn optimal energy trading and battery scheduling strategies purely through trial-and-error interactions with the environment, entirely bypassing the need for explicit transition probability models.

Task 2: Methodology Analysis

(a) ML Technique and Dataset

Technique: The article utilizes a Reinforcement Learning algorithm known as Multi-Agent Deep Deterministic Policy Gradient (MADDPG), which operates within the Actor-Critic framework to handle continuous action spaces.

Dataset & Preparation: The authors used real-world solar irradiance, wind speed, and load consumption data sourced from the National Renewable Energy Laboratory (NREL). The continuous time-series data was discretized into hourly intervals over a full year to formulate the state space (e.g., battery State of Charge, current load, current renewable output) for the Markov Decision Process (MDP).

(b) Mapping to CO 4 and Strengths/Limitations

CO 4 Mapping: Reinforcement Learning.

Methodological Strength: The model-free nature of the RL approach allows it to adapt seamlessly to unpredictable fluctuations in renewable energy without requiring strict physical system modeling.

Methodological Limitation: Deep RL algorithms suffer from sample inefficiency and high variance during the initial exploration phases, leading to massive computational overhead during training before convergence is achieved.

Task 3: Results & Critical Evaluation

(a) Key Results

The authors benchmarked their MADDPG approach against independent Deep Deterministic Policy Gradient (DDPG) and traditional Deep Q-Networks (DQN). The MADDPG algorithm demonstrated superior reward convergence and significantly minimized operational costs.

Algorithm	Average Daily Operational Cost ($)	Reward Convergence Stability	Policy Type
MADDPG (Proposed)	$ 142.30	Highly Stable (Low Variance)	Multi-Agent / Cooperative
Independent DDPG	$ 167.85	Moderate	Single-Agent
DQN	$ 195.40	Unstable (High Variance)	Single-Agent / Discrete

(b) Critical Evaluation

The reported cost reductions are realistic and align with the theoretical benefits of centralized training with decentralized execution. However, a potential bias exists in the evaluation protocol: the model was trained and tested entirely within a simulated NREL data environment, which does not account for sudden hardware failures or extreme anomalous weather events (e.g., blackouts). Additional experiments testing the learned policy on a Hardware-in-the-Loop (HIL) physical microgrid simulator would significantly strengthen the findings and prove real-world robustness.

(c) Real-world Applicability

This RL technique is highly deployable in community microgrids, smart university campuses, and industrial parks looking to optimize their renewable energy footprint. The challenges in scaling to industry include data availability (requiring high-fidelity telemetry from smart meters), vast compute requirements for training multi-agent systems, and ethical/safety concerns. In a physical power grid, a poor exploratory action by an untrained RL agent could cause a localized blackout or battery degradation, necessitating strict "Safe RL" constraints.

Task 4: Reflection & Learning Outcome

(a) Key Insights

- **MDP Formulation is Critical:** Translating physical constraints (like battery capacity limits) into the States, Actions, and Rewards of a Markov Decision Process bridges theoretical RL with applied engineering.

- **Exploration vs. Exploitation Cost:** In physical systems, exploring random actions incurs real financial costs (e.g., wasting energy). This deepens my understanding of why balancing the ϵ-greedy strategy is crucial in Q-learning.
- **Actor-Critic Advantages:** Standard Q-Learning (CO4) handles discrete actions well, but the article demonstrated that Actor-Critic architectures are necessary for continuous action spaces, expanding my perspective on advanced RL models.

(b) Proposed Extension

If I were to extend this research, I would propose incorporating real-time Electric Vehicle (EV) fleet charging dynamics into the multi-agent environment, treating EVs as mobile, dynamic energy storage units.

Feasibility: Highly feasible using existing open-source EV mobility datasets and adjusting the state space of the MDP.

Expected Impact: This enhancement would further reduce microgrid operational costs by absorbing excess renewable generation during peak sun hours and discharging back to the grid during peak evening loads, drastically improving overall grid stability and sustainability.